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Resilience Metrics in Power & Control Systems: Comparative Analysis

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Abstract: Amid the increasing frequency of extreme events, traditional approaches in control and power systems are proving ineffective. There's an essential need for a new perspective to manage the emerging risk landscape and to prioritize the recovery process. Resilience, the ability to cover events across a spectrum from common to catastrophic, extending from pre-event to post-event periods, offers a potential solution. Despite significant research, a universal standard for resilience metrics is lacking. This paper delves into this issue, studying the main existing metrics to pinpoint gaps and suggest future directions for standardized control and power systems resilience assessment.

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Keywords: Power system resilience, Resilience metrics, Extreme events, Reliability

Glossary:

P_0	The initial performance state
P_1	The degraded state
P(t)	The performance function
D	Load demand
LS	Load shedding
VOLL	Value of lost load ($/MWh)

1. INTRODUCTION

Extreme events have increased in frequency in recent decades, presenting challenges to control and power systems. In the literature, terms for these events are used interchangeably based on statistical and historical data (Gholami et al., 2018; Shahidehpour et al., 2016). However, a more appropriate classification distinguishes events as High Impact and Low Frequency (HILF) or extreme events, and Low Impact and High Frequency (LIHF) or common events (Nakiganda et al., 2022). This classification aligns more accurately with defining the frequency of extreme events through statistical and historical data compared to the probability of such events. HILF events encompass extreme weather events and human-made incidents. Notably, data on power outages in the USA from 2018 to 2022 underscore the significant role played by extreme events in more than 50 percent of the outages. According to the World Meteorological Organization, the number of extreme weather events, including hurricanes, ice storms, and floods, has surged fivefold over the last five decades. This increase in extreme events, coupled with various threats such as human errors and cyber-attacks, has put critical infrastructures at risk.

Traditional safety concepts, including reliability, fault-tolerant control (FTC), and robustness design and analysis, have proven effective in assessing system behavior under normal or abnormal conditions following specific probability distributions and common consequences. Nevertheless, these concepts encounter challenges when addressing severely faulted systems and systems behavior during extreme events, especially in scenarios lacking probability distributions. This becomes particularly pronounced when accounting for budgetary constraints and the critical importance of the recovery process. Reliability analysis tools are commonly employed to identify system weaknesses, utilizing redundancy, diversity, and system hardening for improvements. However, their application is more suitable for pre-event periods and common events with predefined probability density distributions. In contrast, robustness is defined as a power system's inherent ability to maintain functionality in changing external conditions, even tolerating slight performance degradation (Carpinelli et al., 2009). While robustness offers a broader scope than reliability by ensuring consistent performance under changing conditions and minor disturbances, it does not adequately address recovery after significant performance declines. Concerning FTC, it stands out as a specialized strategy for handling common faults and maintaining desirable performance and stability properties (Zhang and Jiang, 2008). However, it falls short in addressing unknown consequences and is generally incapable of continuous adaptation and functionality in the face of severe events with large number of faults. Specifically, it is crafted to manage common faults and may prove less effective in handling failures condition and unforeseen events or prolonged periods of stress resulting in performance reduction beyond the minimum desirable (graceful degradation).

Consequently, the need for new design philosophies, analysis methodologies, and tools to fill the gaps has become apparent. This new concept should minimize performance degradation and reduce recovery time by covering vast range of

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disturbances and performance degradations. Resilience, a relatively new concept in the power system industry, stands out by offering a comprehensive approach that addresses HILF and LIHF events and takes into account the pre-, during, and post-event conditions. In fact, reliability is the fundamental concept, forming the base. Robustness builds upon reliability, extending its scope by ensuring consistent performance under varying conditions and minor disturbances. The FTC takes the next step, proactively handling and mitigating the effects of common faults within a system by maintaining the availability and stability of the system during events. Resilience, however, transcends these individual concepts. It not only encompasses reliability, robustness, and fault-tolerant control but also extends the paradigm to effectively address major disruptions, including both known and unknown consequences. Its comprehensive nature ensures a prompt recovery and adaptive response to diverse challenges (Younesi et al., 2022). Despite significant progress, a consensus on the definition and standard evaluation techniques for power system resilience is lacking. While some studies have attempted to use other safety metrics to analyze the impact of extreme events on power systems, these metrics are passive. Resilience metrics should incorporate the dynamic and spatiotemporal behaviors of systems in the presence of extreme events (Gholami et al., 2018; Panteli et al., 2017).

While valuable studies have been conducted to compare resilience metrics, they often fall short in identifying gaps in detail and across different scenarios (Afzal et al., 2020; Gholami et al., 2018). This paper critically assesses the effectiveness of developed metrics for resilience analysis by conceptual examples, highlights their limitations, and provides insights for future research. The structure of the paper is as follows: Section 2 delves into the primary metrics for control and power systems resilience assessment, Section 3 utilizes conceptual examples to illustrate the limitations of the metrics, and Section 4 identifies current gaps and suggests potential directions for future investigations.

2. RESILIENCE METRICS

In recent decades, the electrical power industry has increasingly embraced the concept of resilience. Numerous working groups were formed, with specialists from various organizations collaborating closely to establish a shared definition of resilience. For example, the IEEE PES Task Force introduced resilience as a relatively new concept in the power system domain as “Power system resilience is the ability to limit the extent, system impact, and duration of degradation in order to sustain critical services following an extraordinary event. Key enablers for a resilient response include the capacity to anticipate, absorb, rapidly recover from, adapt to, and learn from such an event. Extraordinary events for the power system may be caused by natural threats, accidents, equipment failures, and deliberate physical or cyber-attacks” (Stanković et al., 2023). It is worth noting that the power industry is not alone; various other industries exploring resilience design concept. Resilience definitions in various fields including civil, computer science, climatology, organization and society, infrastructure systems, and network systems has been discussing (Younesi et al., 2022).

The core requirement for assessing resilience is the establishment of a set of measures linked to specific attributes. These metrics are commonly known as resilience indexes, indicators, or measures. Resilience, being a multidimensional concept with varying considerations in different situations, cannot be adequately captured by a single measure. These metrics serve various purposes, such as planning preventive measures to mitigate the adverse effects of events and identifying strengths and weaknesses within a system. By reinforcing strengths and addressing identified weaknesses, overall system resilience can be strengthened. General principles have been suggested for resilience metrics (Panteli and Mancarella, 2015; Vugrin et al., 2017):

- Prioritization of HILF events: Metrics should prioritize High Impact Low Frequency (HILF) events and their consequences in their development.
- Consideration of uncertainties and interdependencies: Metrics should accurately reflect the uncertainties within systems and acknowledge the interdependencies among critical infrastructures.
- Spatiotemporal, consistent, and performance-based: Resilience metrics should incorporate spatiotemporal aspects, maintain consistency, and be centered on system performance.

Metrics should also encompass factors like uncertainties and recovery/restoration time. In (Raoufi et al., 2020), fifteen attributes for resilience metrics are suggested, emphasizing their alignment with the resilience definition, ability to differentiate between resilient and non-resilient systems, and applicability to both operational and planning phases. Various studies have focused on identifying essential features for resilience metrics in power systems, accounting for aspects like the roles of active and reactive power in frequency and voltage stability, as well as the impact of Distributed Energy Resources (DERs) and flexible assets within the network (McJunkin and Rieger, 2017). Additionally, the level at which resilience metrics are applied can vary; they can pertain to either individual components or entire system levels. A conceptual framework for sorting resilience metrics in power systems has been presented in (Mahdavi Tabatabaei et al., 2019; Raoufi et al., 2020) including performance-based and non-performance-based. Performance-based metrics are calculated based on a system's performance and quantify the degree to which a system exhibits resilience.

In contrast, non-performance-based metrics evaluate the impact of events on the economy, society, and the environment, and they are often assessed using on-site visits and questionnaires, involving binary "Yes" or "No" responses to identify critical attributes in the systems. Some studies have considered non-performance-based metrics, although many have primarily focused on performance-based metrics. Nevertheless, while these classifications serve as general guidelines applicable to various resilience concepts, a more specific and detailed categorization is essential.

Table 1. The summary of existing main resilience metrics

No.	Metric	Type	Characteristics	Formulation	Reference
1	Loss of Load Frequency (LOLF)	Reliability-based	Using reliability metrics in analyzing resilience	$\frac{\text{Total number of loss of loads}}{\text{Years}}$	(Mujjuni et al., 2023)
2	Loss of Load Expectation (LOLE)			$\frac{\text{Total duration of loss of loads (hour)}}{\text{Years}}$	
3	Energy Not Supplied (ENS)			$\int (P_0 - P(t))dt$	
4	F or ϕ	Trend-based	Define the performance level reduction immediately after the event	The slope of performance reduction from initial state (P_0) to degraded state (P_1): $(\frac{P_1-P_0}{T_1})$	(Panteli et al., 2017)
5	L or Λ		Define the magnitude of damage	$(P_0 - P_1)$	
6	E		Define the duration of damage	The period remains in the degraded state (P_1): $(T_2 - T_1)$	
7	P or Π		Define the speed and quality of the recovery process	The slope of performance recovery from degraded state (P_1) to initial state (P_0): $(\frac{P_0-P_1}{T_3-T_2})$	
8	Load-shedding investment cost	Optimization -based	Study the resourcefulness attribute in the systems	$\Delta C_{LS} = VOLL * (D - LS) = VOLL * ENS$	(Abbasi et al., 2019)
9	R	Performance-based	Defined based on the area under the resilience curve	$R = \frac{\int P(t)dt}{\int P_0 dt}$	(Afzal et al., 2020; Gholami et al., 2018)

A comprehensive overview of well-established resilience metrics is provided in Table 1, categorized according to their distinctive features. These metrics can be methodically grouped into four primary categories: reliability-based metrics, trend-based metrics, optimization-based metrics, and performance-based metrics. Reliability-based metrics, while occasionally adapted for resilience assessments, exhibit limitations in capturing the dynamic and time-varying behavior of systems under extreme events. This observation holds when applying metrics from other safety concepts, such as fault-tolerant control and robustness, for resilience analysis. For instance, when facing a sophisticated cyber-attack targeting multiple components of the power system, traditional fault-tolerant control metrics tailored for single-fault scenarios may encounter challenges in comprehensively assessing the intricacies of a coordinated attack that impacts various components simultaneously. This limitation arises because these metrics primarily concentrate on aspects such as detection and isolation time (measuring how rapidly the system identifies and isolates faulty components) and availability, which indicates the probability of the system being operational at any given time. In contrast, resilience metrics must portray the dynamic and spatiotemporal aspects of a system's behaviour, encompassing its state before and after events with any degree of degradation. Trend-based metrics, within the second category, offer a more precise representation of a system's behavior. For instance, four trend-based metrics, F, L, E, and P, delineate parameters such as the reduction in performance levels, the speed of reduction, the extent of post-event recovery, and the pace of the recovery process by simplifying the performance curve to a trapezoidal shape, as illustrated in Figure 1. Despite their potential, these metrics face challenges in achieving universal acceptance, and

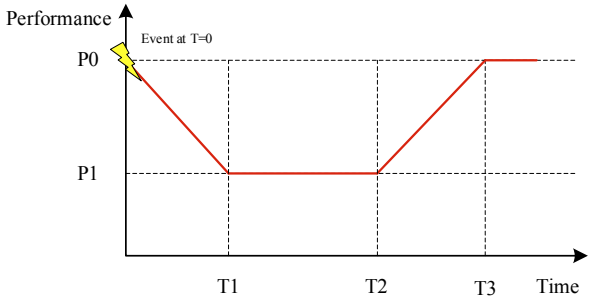


Figure 1. Resilience/Performance curve trapezoid.

their determinants in identical conditions are unclear (Panteli et al., 2017). Optimization-based metrics, in the third category, are centered on identifying the most cost-effective restoration processes by optimizing operational and topological conditions. While these metrics provide valuable insights, they are inherently static and do not capture the dynamic behaviour of systems in the post-event phase. Nevertheless, they often rely on simplifying assumptions, rendering the implementation of such optimized solutions challenging, and sometimes infeasible in practical post-event scenarios. While valuable, the fourth category, performance-based metrics, may only encompass some necessary resilience features. For instance, two normalized resilience metrics were developed based on performance curves (Gholami et al., 2018). Although normalization aids in comparing utilities, these metrics might not capture a system's dynamic behavior under harsh conditions. Both metrics rely on calculating the area under a performance curve, yet in resilience assessment, it's essential to consider both the rate of performance deterioration and the minimum performance level.

3. COMPARATIVE ANALYSIS

This section utilizes conceptual examples to assess the metrics discussed in Section 2. In the context of power system performance, we can consider factors such as the number of damaged lines, shaded loads, the number of connected sources, etc. For the purposes of this study, demand power (the amount of electricity consumers need from the grid) is considered for the performance curve. The conceptual behavior of four distinct power systems under an extreme event occurring at t = 0 is depicted in Figure. 2. The results of resilience metrics are presented in Table 2. The observation period for calculating reliability-based metrics is based on a single event, with just one system failure due to the extreme event occurring annually. The VOLL is also considered at 1000 $/MWh for the optimization-based metric calculations.

The initial observation highlights that the results do not distinctly reveal the resilience level for each system or facilitate precise comparisons among them. In this context, all the reliability-based metrics and the optimization-based metric produce identical results, suggesting that, from the standpoint of these metrics, all four systems exhibit a similar level of resilience. While reliability and resilience share commonalities, they also possess distinct differences. Reliability can be defined as the probability that a system performs its intended functions in a specific time and under certain circumstances. In essence, reliability offers an estimation based on the current state of knowledge (in pre-event phase). In contrast, resilience demonstrates dynamic behavior that cannot be adequately captured by the formulation and metrics of reliability. It emphasizes the importance of understanding the system's condition and performance level following an event, recognizing that resilience evolves over time. Moreover, the reliability-based metrics are incapable of considering the recovery process and utilizing the available resources optimally in the post-event phase. In the case of trend-based metrics, the significance and prioritization of the FLEP remain unspecified. System A

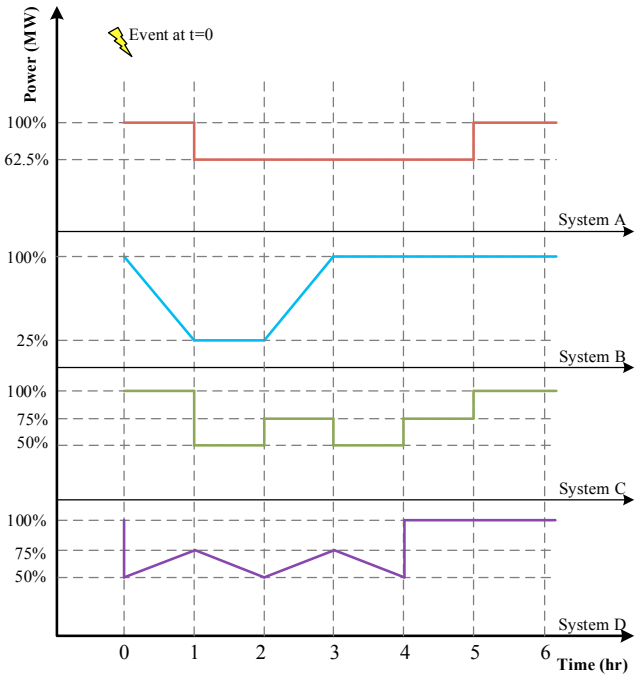


Figure. 2. The four conceptual case studies.

exhibits an abrupt performance reduction, remaining in the post-event phase for an extended period compared to System B. Conversely, System B demonstrates a slower recovery process but experiences more substantial performance reduction, up to 75%. Determining the preferred system from a resilience perspective remains ambiguous. Moreover, these metrics necessitate the simplification of the performance curve into a trapezoidal shape. However, in cases such as Systems C and D, there are multiple ways to perform this simplification, and the results may overlook vital aspects of studying systems behavior under extreme events. Consequently, these metrics fall short in accounting for the total duration of the recovery process, from the initiation of performance reduction to the recovery of normal conditions (the broader base of the trapezoid). They solely consider the time systems spend within the shorter base of the trapezoid.

Similar to the reliability-based metrics, performance-based metrics rely on integration to represent the cumulative resilience metric. However, these metrics have significant limitations as they fail to capture the comprehensive essence of resilience. Their primary focus is on quantifying the cumulative impact of deviations from nominal values, thus neglecting certain crucial resilience characteristics. For instance, none of these metrics account for the duration of resistance from the initiation of an event until the onset of performance reduction. Consider Systems A and D, which are regarded as equal in most metrics. Nonetheless, System A resists performance degradation for 1 hour before experiencing a decline. This resistance period provides operators with valuable time to implement effective mitigation measures. Furthermore, these metrics do not address the stability of systems during the recovery process. System D exhibits greater instability with significant up-and-down variations, which, in practice, makes the recovery process

Table 2. The resilience metrics results

No.	Metric	System A	System B	System C	System D
1	Loss of Load Frequency (1/year)	1	1	1	1
2	Loss of Load Expectation (h/year)	4	3	4	4
3	Energy Not Supplied (MWh)	1.5	1.5	1.5	1.5
4	F or ϕ	$-\infty$	-0.75	N/A	N/A
5	L or Λ	0.625	0.25	N/A	N/A
6	E	4	1	N/A	N/A
7	P or Π	$+\infty$	+0.75	N/A	N/A
8	Load-shedding investment cost (USD)	1500	1500	1500	1500
9	R	0.625	0.5	0.625	0.625

Table 3. The definition of main characteristics of resilience

Characteristic	Definition	Reference
Resistance	The characteristic embodies a multifaceted objective. Initially, its primary aim is to ensure reliability by consistently fulfilling its intended function and exhibiting strong resistance to adverse events. Additionally, it endeavors to achieve robustness, enabling the system to sustain functionality and performance even when confronted with minor degradations. Furthermore, when facing severe degradation, the overarching goal is to withstand the initial impact of an event, managing the decline in performance gradually rather than experiencing a sudden and sharp drop. This gradual reduction is paramount in cases where system reliability and robustness may falter. It not only helps prevent physical damage to the equipment but also provides operators with valuable time to enact effective mitigation measures.	(Afzal et al., 2020; Stanković et al., 2023)
Absorption	The system's ability to minimize damages incurred due to events is a critical attribute. This characteristic quantifies the average reduction in performance and evaluates the system's capability and resilience to absorb the damages resulting from the event.	(Bajwa et al., 2019; Jufri et al., 2019)
Recovery	This attribute reflects the system's ability to swiftly return to its normal performance. It encompasses both operational restoration and infrastructural recovery.	(Afzal et al., 2020; Stanković et al., 2023)
Stability	The stability characteristic refers to the system's capacity to maintain stability and robustness after an event. Successful recovery depends on the system's ability to sustain stability during the restoration process. High fluctuations in performance can significantly hamper the recovery process.	(Younesi et al., 2022)
Adaptivity	This attribute involves leveraging learning and improvement mechanisms derived from past disruptions. It encompasses the system's ability to extract insights from historical data, past experiences, and simulations, enhancing responsiveness to future scenarios.	(Panteli and Mancarella, 2015)

more challenging compared to System A. It introduces severe mechanical stresses to the equipment, complicating the overall recovery.

In summary, resilience metrics should be logically aligned with the fundamental characteristics of resilience. According to established resilience definitions (Chaudry et al., 2011; Panteli and Mancarella, 2015; Stankovic and Tomsovic, 2018), the main attributes of resilience encompass anticipation, resistance, absorption, rapid recovery, and adaptivity. However, the concept of resilience extends to severe and unprecedented events, making anticipation less suitable for inclusion within resilience definitions. Moreover, a resilient system must preserve essential attributes like voltage and frequency stability as the recovery process initiates to prevent further performance degradation. Hence, appropriate resilience characteristics should include resistance, absorption, recovery, stability, and adaptivity, as defined in Table 3.

4. CONCLUSION

This study aimed to briefly underscore the significance of resilience analysis and provide an overview of the primary developed resilience metrics. While employing more realistic case studies might offer greater precision, the conceptual examples effectively highlight the limitations of the metrics. The results collectively reveal that the existing metrics fall short in facilitating accurate system comparisons from a resilience perspective and fail to comprehensively reflect all facets of resilience. Moreover, further comprehensive investigations are needed to establish clear distinctions in the

definitions of resilience as compared to similar concepts such as reliability and fault-tolerance. This distinction, supported by experimental data and simulation results, is essential. Precisely defining the boundaries of these concepts and understanding their respective application domains will significantly enhance the design of resilient power systems. Last but not least, it is crucial to consider the interdependencies among infrastructures in the case of resilience analysis under the effect of extreme events.

REFERENCES

Abbasi, S., Barati, M., Lim, G.J., 2019. A Parallel Sectionalized Restoration Scheme for Resilient Smart Grid Systems. IEEE Trans. Smart Grid 10, 1660–1670. https://doi.org/10.1109/TSG.2017.2775523

Afzal, S., Mokhlis, H., Illias, H.A., Mansor, N.N., Shareef, H., 2020. State-of-the-art review on power system resilience and assessment techniques. IET Generation, Transmission & Distribution 14, 6107–6121. https://doi.org/10.1049/iet-gtd.2020.0531

Bajwa, A.A., Mokhlis, H., Mekhilef, S., Mubin, M., 2019. Enhancing power system resilience leveraging microgrids: A review. Journal of Renewable and Sustainable Energy 11, 035503. https://doi.org/10.1063/1.5066264

Carpinelli, G., Di Perna, C., Caramia, P., Varilone, P., Verde, P., 2009. Methods for Assessing the Robustness of Electrical Power Systems Against Voltage Dips. IEEE Trans. Power Delivery 24, 43–51. https://doi.org/10.1109/TPWRD.2008.2002991

Chaudry, M., Ekins, P., Ramachandran, K., Shakoor, A., Skea, J., Strbac, G., Wang, X., Whitaker, J., 2011. Building a Resilient UK Energy System: Research Report. UK Energy Research Centre (UKERC), London, UK.

Gholami, A., Shekari, T., Amirioun, M.H., Aminifar, F., Amini, M.H., Sargolzaei, A., 2018. Toward a Consensus on the Definition and Taxonomy of Power System Resilience. IEEE Access 6, 32035–32053. https://doi.org/10.1109/ACCESS.2018.2845378

Jufri, F.H., Widiputra, V., Jung, J., 2019. State-of-the-art review on power grid resilience to extreme weather events: Definitions, frameworks, quantitative assessment methodologies, and enhancement strategies. Applied Energy 239, 1049–1065. https://doi.org/10.1016/j.apenergy.2019.02.017

Mahdavi Tabatabaei, N., Najafi Ravadanegh, S., Bizon, N. (Eds.), 2019. Power Systems Resilience: Modeling, Analysis and Practice, Power Systems. Springer International Publishing, Cham. https://doi.org/10.1007/978-3-319-94442-5

McJunkin, T.R., Rieger, C.G., 2017. Electricity distribution system resilient control system metrics, in: 2017 Resilience Week (RWS). Presented at the 2017 Resilience Week (RWS), IEEE, Wilmington, DE, USA, pp. 103–112. https://doi.org/10.1109/RWEEK.2017.8088656

Mujjuni, F., Betts, T.R., Blanchard, R.E., 2023. Evaluation of Power Systems Resilience to Extreme Weather Events: A Review of Methods and Assumptions. IEEE Access 11, 87279–87296. https://doi.org/10.1109/ACCESS.2023.3304643

Nakiganda, A.M., Dehghan, S., Markovic, U., Hug, G., Aristidou, P., 2022. A Stochastic-Robust Approach for Resilient Microgrid Investment Planning Under Static and Transient Islanding Security Constraints. IEEE Trans. Smart Grid 13, 1774–1788. https://doi.org/10.1109/TSG.2022.3146193

Panteli, M., Mancarella, P., 2015. The Grid: Stronger, Bigger, Smarter? IEEE Power and Energy Mag. 13, 58–66. https://doi.org/10.1109/MPE.2015.2397334

Panteli, M., Mancarella, P., Trakas, D.N., Kyriakides, E., Hatziargyriou, N.D., 2017. Metrics and Quantification of Operational and Infrastructure Resilience in Power Systems. IEEE Trans. Power Syst. 32, 4732–4742. https://doi.org/10.1109/TPWRS.2017.2664141

Raoufi, H., Vahidinasab, V., Mehran, K., 2020. Power Systems Resilience Metrics: A Comprehensive Review of Challenges and Outlook. Sustainability 12, 9698. https://doi.org/10.3390/su12229698

Shahidehpour, M., liu, xindong, Li, Z., Cao, Y., 2016. Microgrids for Enhancing the Power Grid Resilience in Extreme Conditions. IEEE Trans. Smart Grid 1–1. https://doi.org/10.1109/TSG.2016.2579999

Stankovic, A., Tomsovic, K., 2018. The Definition and Quantification of Resilience (No. PES-TR65). IEEE PES Industry Technical Support Task Force.

Stanković, A.M., Tomsovic, K.L., De Caro, F., Braun, M., Chow, J.H., Čukalevski, N., Dobson, I., Eto, J., Fink, B., Hachmann, C., Hill, D., Ji, C., Kavicky, J.A., Levi, V., Liu, C.-C., Mili, L., Moreno, R., Panteli, M., Petit, F.D., Sansavini, G., Singh, C., Srivastava, A.K., Strunz, K., Sun, H., Xu, Y., Zhao, S., 2023. Methods for Analysis and Quantification of Power System Resilience. IEEE Trans. Power Syst. 38, 4774–4787. https://doi.org/10.1109/TPWRS.2022.3212688

Vugrin, E., Castillo, A., Silva-Monroy, C., 2017. Resilience Metrics for the Electric Power System: A Performance-Based Approach. (No. SAND2017-1493, 1367499, 654236). https://doi.org/10.2172/1367499

Younesi, A., Shayeghi, H., Wang, Z., Siano, P., Mehrizi-Sani, A., Safari, A., 2022. Trends in modern power systems resilience: State-of-the-art review. Renewable and Sustainable Energy Reviews 162, 112397. https://doi.org/10.1016/j.rser.2022.112397

Zhang, Y., Jiang, J., 2008. Bibliographical review on reconfigurable fault-tolerant control systems. Annual Reviews in Control 32, 229–252. https://doi.org/10.1016/j.arcontrol.2008.03.008